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SYSTEMATIC REVIEW

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# Beyond binary comparisons: a Bayesian dose-response meta-analysis of exercise on executive function in children and adolescents with ADHD

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**OBJECTIVE:** This study aimed to systematically evaluate the acute and long-term effects of exercise interventions on executive function in children and adolescents with attention-deficit/hyperactivity disorder (ADHD), using Bayesian dose–response modelling to identify optimal dose ranges and modality-specific effects.
**METHODS:** A systematic search of five major databases (PubMed, Web of Science, Embase, Cochrane Library, and APA PsycInfo) was conducted up to March 2025. Thirty-three eligible studies were included, comprising 10 acute and 23 long-term exercise intervention trials. Bayesian non-linear dose–response models were applied to determine optimal doses for three executive function domains: cognitive flexibility, working memory, and inhibitory control.
**RESULTS:** Exercise interventions significantly improved executive function in youth with ADHD and showed clear dose-dependent patterns. In acute interventions, optimal doses were 270 METs for cognitive flexibility, 170 METs for working memory, and 130 METs for inhibitory control. In long-term interventions, optimal weekly doses were 1100, 1300, and 2500 METs, respectively. Different exercise modalities yielded varying effects across executive domains.
**CONCLUSION:** Exercise benefits executive function in children and adolescents with ADHD, with effects influenced by both dose and modality. This study proposes a "dose–domain–modality" framework, offering empirical support for individualized, precision-based exercise prescriptions.

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**IMPACT:**

- This is the first systematic review and dose–response meta-analysis quantifying acute and long-term effects of exercise on executive function in ADHD youth, identifying domain-specific optimal doses for cognitive flexibility, working memory, and inhibitory control across 33 studies.
- By integrating Bayesian modeling and network meta-analytic methods, the study introduces a "dose–domain–modality" framework, offering novel empirical support for individualized, function-targeted exercise prescriptions.
- It moves beyond merely showing effectiveness or a single "optimal" dose, emphasizing precise prescriptions tailored to executive function domains for targeted cognitive enhancement.

## INTRODUCTION

Attention-deficit hyperactivity disorder (ADHD) is a neurodevelopmental disorder that originates in childhood, typically characterized by persistent difficulties in attention maintenance, impulsive behaviors, and hyperactivity, often exceeding the developmental norms for age-matched peers.[1] Epidemiological data indicate that over 5% of children and adolescents worldwide are diagnosed with ADHD,[2] and symptoms persist into adulthood in a substantial proportion, reflecting notable developmental continuity.[3] Previous studies have demonstrated widespread deficits in executive functions among individuals with ADHD, particularly in unstable working memory, impaired cognitive flexibility, and abnormal inhibitory control, with neuroimaging studies further supporting a neural mechanism involving impaired prefrontal cortical functioning.[4] Without effective early intervention, such functional deficits may have persistent negative impacts on academic performance, psychological development, and social adaptability, increasing the risk of academic underachievement, behavioral issues, and social exclusion.[5] Considerable evidence supports exercise interventions as an effective means of enhancing executive functions—including working memory, cognitive flexibility, and inhibitory control-in ADHD patients. Studies indicate that even a single exercise session can increase neurotransmitter levels such as dopamine, serotonin, and endorphins, thus promoting neural

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plasticity and subsequently improving executive function.[6] Meanwhile, long-term interventions accumulate beneficial effects, yielding more sustained improvements.[7] Furthermore, there is a positive correlation between exercise training and improved executive function.[8] Compared to pharmacological approaches, exercise interventions offer advantages such as fewer side effects-avoiding risks associated with medication dependence and cardiovascular responses[9–11]-alongside lower costs, higher acceptability, and greater adaptability, making them highly promising for clinical application.[12]

Extensive systematic reviews and meta-analyses have established that exercise interventions can improve executive function in children and adolescents with ADHD.[13–16] However, these studies typically treat exercise as a simple binary variable—merely recording whether an intervention was implemented—without systematically incorporating critical parameters such as duration (weeks), frequency (sessions per week), session length, or intensity. Even those employing network meta-analyses to compare different exercise programs generally rely on superficial categorical classifications, ranking interventions by nominal type while overlooking substantial differences in duration, frequency, and session length across trials. As a result, what appear to be rigorous comparisons may in fact juxtapose programs with vastly different total exposures-for example, directly comparing an 18-week intervention to one lasting only 4 weeks, attributing all observed differences solely to intervention type. This markedly overstates the supposed inherent superiority of certain modalities and obscures the fundamental role of total exercise dose in shaping outcomes. While some analyses have explored the impact of frequency or overall intervention time through single-factor linear meta-regressions, these studies typically lack a standardized framework that integrates intensity, frequency, session length, and duration into a unified metric of total dose. Few have modeled non-linear dose–response relationships within a consistent analytical structure. Consequently, key questions remain unanswered: does increasing total exercise dose lead to diminishing returns or even saturation in executive function improvements? Do acute and long-term interventions exhibit different efficacy thresholds under comparable cumulative loads? These long-neglected issues have created a clear gap in the literature and directly constrain the movement toward individualized applications of exercise interventions in practice. In the absence of robust, quantitatively defined dose recommendations, current programs often rely on empirical heuristics or broad general guidelines, making it difficult to achieve truly personalized, precise interventions for addressing executive dysfunction in youth with ADHD.

To address these fundamental shortcomings, this study systematically synthesized current experimental data on exercise interventions aimed at improving executive function in children and adolescents with ADHD. Unlike previous research that primarily treated exercise as a simple binary variable-whether an intervention occurred-or compared different exercise modalities without accounting for critical discrepancies in intensity, frequency, session duration, and overall length, we integrated these parameters into a unified, standardized total dose metric (expressed as METs-min/week). This study employed a Bayesian dose–response meta-analytic approach, extending the latest applications of this methodology in the field of exercise and cognition,[17] we characterized the non-linear relationships for acute and long-term interventions within the same analytic framework, identifying marginal benefits, optimal dose ranges, and potential saturation points. This approach not only corrects for biases introduced by past studies that classified interventions based solely on nominal type despite vast differences in total exposure, but also fills a major methodological gap by incorporating multi-dimensional dose information into non-linear modeling. Our findings thus provide a robust quantitative foundation for individualized, evidence-based exercise prescriptions and advance both theoretical understanding and practical strategies for addressing executive dysfunction in youth with ADHD.

## METHODOLOGY

### Search strategy

This study adhered to international standards for systematic reviews and meta-analyses, following the PRISMA guidelines to structure the research process,[18] with prior registration on the PROSPERO platform (registration number: CRD420251022059). A comprehensive literature search was conducted across five major databases—Cochrane Library, Web of Science, PubMed, Embase, and APA PsycInfo—covering records from their inception to 20 March 2025. The search strategy integrated keywords, abstracts, and subject headings, tailored to the syntax of each database, and imposed no language restrictions. In addition, we manually screened the reference lists of relevant systematic reviews published in the past decade to ensure the comprehensiveness and representativeness of the included studies. Two reviewers independently screened the records within a unified framework, and a third reviewer verified all inclusion decisions to ensure consistency and completeness. Full search strategies are detailed in Supplementary File 1.

### Study selection and eligibility criteria

This review aimed to systematically evaluate the effects of different types and dosages of exercise interventions on executive function in children and adolescents diagnosed with ADHD. The inclusion criteria were as follows:

*Participants*. Children and adolescents formally diagnosed with ADHD by qualified professionals based on established diagnostic criteria, such as the Diagnostic and Statistical Manual of Mental Disorders (DSM) or other commonly used clinical scales.[19] To ensure specificity of the intervention effects, we excluded studies involving participants with comorbid autism spectrum disorder, Tourette syndrome, epilepsy, or other neurodevelopmental conditions. Studies with control groups that included non-ADHD individuals were also excluded to improve sample homogeneity. No restrictions were placed on ADHD subtypes, sex, family background, socioeconomic status, cognitive ability, ethnicity, or country of origin, allowing for broader generalizability of the findings.

*Interventions*. All types of exercise interventions were eligible, including various formats, frequencies, intensities, durations, and modes such as ball games, running, resistance training, and aerobic games. Interventions could be single-session or multi-session and may involve structured training programs of differing designs.

*Comparators*. Acceptable control conditions included usual care, health education, or no additional intervention. To facilitate the network meta-analysis framework, certain exercise modalities were treated as comparators in studies designed to enable indirect comparisons between different interventions.

*Outcomes*. Studies were required to report objective measures of executive function in ADHD populations, specifically targeting three core domains: inhibitory control, working memory, and cognitive flexibility. Commonly used assessment tools included the Continuous Performance Test, Trail Making Test (TMT), Digit Span Task (DST), and Stroop Test.

*Study design*. We prioritized randomized controlled trials (RCTs), including both parallel-group and crossover designs. For crossover trials, participants were required to receive the interventions in a randomized order to reduce inter-individual variability, a

particularly relevant design for evaluating the acute effects of exercise on executive functioning.[20] Only peer-reviewed published articles were considered. In cases where multiple studies used the same sample and reported outcomes at different time points, only the latest available data were included in the analysis.

To further enhance study quality and data comparability, we applied the following exclusion criteria: studies with unavailable raw data, even after two follow-up emails to the first or corresponding author (with a 7-day interval between attempts); studies that included executive function measures but primarily focused on other ADHD-related domains such as mental health, social skills, or sleep; studies with low methodological quality, defined as a revised PEDro score below 4;[21,22] study protocols; and duplicate publications, in which case only the version with the most complete information was retained.

### Data extraction

In this study, two reviewers independently screened eligible studies and extracted data according to predefined inclusion and exclusion criteria. In cases of disagreement, consensus was first sought through discussion; if no resolution could be reached, a third reviewer-who was not involved in the initial screening-made the final decision, ensuring independence and objectivity in study selection. Extracted data covered the following domains: basic study information (first author, year of publication, country, study design), participant demographics and clinical characteristics (sample size, sex distribution, mean age, ADHD subtype classification, diagnostic criteria, medication status), exercise-related variables (type of intervention, intensity, frequency, total duration, session length, associated metabolic equivalents [METs], and corresponding MET codes), and executive function outcomes before and after intervention.

For studies that did not directly report means and standard deviations, data were estimated using recommended statistical methods based on available parameters such as sample size, medians, ranges, and interquartile ranges.[23,24] When only standard errors, confidence intervals, or interquartile ranges were provided, standard conversion formulas were applied to ensure consistency in data format for statistical synthesis.[25] For studies reporting results exclusively in graphical form, data were extracted using digital image analysis tools when original numerical data could not be obtained from the authors, thereby maximizing data recovery accuracy.[26,27] In studies that employed more than one measure of executive function, only one was extracted to avoid overweighting individual studies in the meta-analysis. In such cases, behavioral measures were prioritized over questionnaire-based assessments, based on their greater objectivity.

### Methodological quality assessment

Two reviewers independently assessed the methodological quality of all included studies using the PEDro scale. Any discrepancies in scoring were resolved through discussion; if disagreement persisted, a third reviewer adjudicated to reach a consensus. The PEDro scale comprises 11 items designed to evaluate the internal validity and statistical integrity of clinical trials. The first item, which addresses external validity, is not included in the final score; the remaining 10 items yield a total score ranging from 0 to 10. According to the preset thresholds for this study, trials scoring below 4 were considered low quality and excluded. Studies with scores of 4–5 were classified as fair quality, 6–7 as moderate quality, and scores of 8 or higher were considered high quality.[25,28] The PEDro scale is widely used in the fields of physiotherapy and exercise-based interventions, with demonstrated reliability and discriminative power.

### Statistical analysis

A network meta-regression model within a Bayesian random-effects framework (MBNMA) was employed to examine the nonlinear dose–response relationship between exercise exposure and executive function outcomes in children and adolescents with ADHD.[29] Prior to formal modeling, the analyzability of the network structure was confirmed, with particular attention to connectivity, transitivity, and consistency assumptions.[30] Intervention effects were estimated using standardized mean differences (SMD, Hedges' g) with 95% credible intervals (CrIs), enabling a quantitative comparison of different exercise types and dosages.[31] Dose–response curves were initially constructed based on observed trends, followed by the fitting of several nonlinear functions—including Emax models, restricted cubic splines, quadratic polynomials, and non-monotonic growth curves—to capture underlying relationships.[32] Competing models were compared using the Deviance Information Criterion (DIC), model complexity, residual distribution, and heterogeneity metrics to identify the best-fitting function.[33] The quadratic function consistently demonstrated superior performance across fit indices and was therefore selected as the final model for dose–response estimation. All analyses were conducted using R (v4.4.2), with the "MBNMAdose" package used for model construction and estimation, and "ggplot2" for curve visualization.

Exercise interventions were first categorized by type, based on their specific content, and then quantified using METs derived from the Compendium of Energy Expenditures for Youth.[34] For acute exercise interventions, energy expenditure was calculated by multiplying the MET value by the duration of a single session; for sustained interventions, weekly energy expenditure was computed as the product of session duration and frequency, yielding a metric expressed in MET-minutes per week.[35] To facilitate the network meta-analysis and enhance statistical stability, continuous energy expenditure data were grouped into discrete dose categories. Specifically, for long-term interventions, categories ranged from control to 100–1500 MET-min/week; for acute interventions, categories ranged from control to 50–300 MET-min per session. This categorization not only improved comparability across studies but also served as a critical preprocessing step in network model construction.[36] Given the substantial variability in energy expenditure across studies, ungrouped continuous data could lead to biased estimates or disproportionate study weights, particularly when sample sizes are sparse or highly clustered at certain dose levels; the standardized grouping strategy thus ensured greater model robustness and interpretability.

### Quality of evidence assessment

Two reviewers independently assessed the quality of evidence from included studies. In cases of disagreement, a third author-independent of the initial assessment-facilitated consensus through discussion. Evidence quality was evaluated using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach,[37] which considers five domains: study limitations, inconsistency, indirectness, imprecision, and publication bias. For each primary outcome, the overall quality of evidence was categorized into four levels according to GRADE criteria: high, moderate, low, or very low.

## RESULTS

### Literature search process

As of 20 March 2025, a total of 5650 records were retrieved from Cochrane Library, Web of Science, PubMed, Embase, and APA PsycInfo. After removing duplicates, 4218 records underwent title and abstract screening, with 4023 excluded. The remaining 195 articles were assessed in full text, of which 162 were excluded due to non-conformity with inclusion criteria regarding intervention, population, study design, or outcomes. Ultimately, 33 studies were included in the meta-analysis. To enhance the comprehensiveness of the search, an additional 184 records were identified through

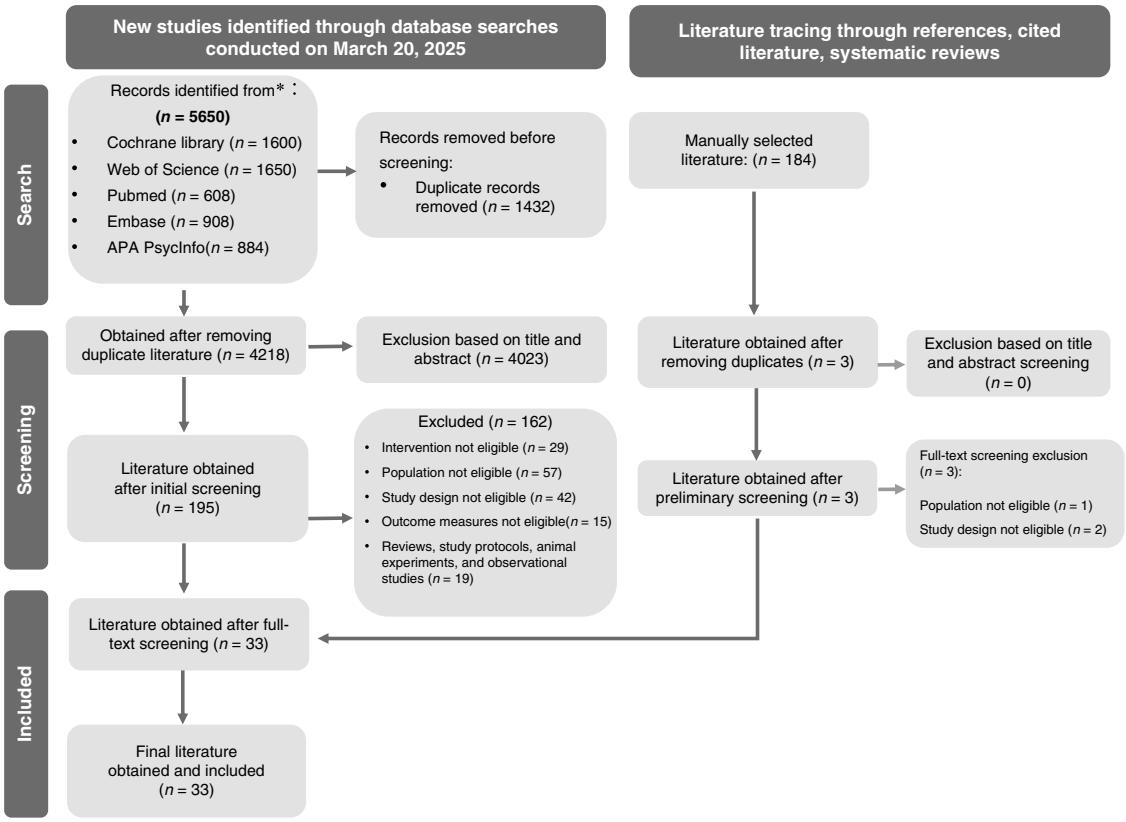



**Fig. 1** Flow diagram of literature screening and inclusion.

backward citation tracking and reference screening of systematic reviews. After deduplication and preliminary review, 3 articles were retained; however, none were eligible for final inclusion following full-text assessment (Fig. 1).

### Study characteristics

This meta-analysis included 33 studies, comprising both acute and long-term exercise interventions. Ten studies involved single-session (acute) interventions, with no reporting of intervention period or weekly frequency (denoted as "/"), while the remaining 23 studies examined sustained interventions. The dose–response relationships were analyzed separately for acute and long-term interventions. Participants were predominantly children and adolescents aged 8–14 years. Most studies applied DSM-IV diagnostic criteria, ensuring clinical consistency across samples (Supplemental Table S1).

All interventions followed structured exercise protocols. Running was the most commonly employed modality ($n = 9$), followed by cycling, martial arts, swimming, and virtual reality-based training; several studies also incorporated combined cognitive-physical training. Exercise intensity was quantified using METs, ranging from 8.4 to 13.1 METs/min. Frequency varied from 1 to 3 sessions per week, with program durations spanning 6–20 weeks and session lengths ranging from 30 to 90 min. Approximately 36% of studies reported concurrent pharmacological treatment (e.g., methylphenidate), although existing evidence suggests that medication did not modify the effect of exercise on executive function in ADHD populations. Other studies did not report or did not include pharmacological interventions. Most of the included studies employed passive control conditions, typically involving quiet rest or the continuation of usual daily activities without any additional intervention. This approach helps isolate the independent effects of exercise on executive function by minimizing potential confounding from extra cognitive or social engagement, thereby enabling a clearer comparison between exercise interventions and no-intervention scenarios.

Outcomes targeted the three core domains of executive function in ADHD: working memory, inhibitory control, and cognitive flexibility. Assessment tools included the Stroop Test (ST, $n = 10$), Conners' Parent Rating Scale, Wisconsin Card Sorting Test, Finger Tapping Task, TMT, and DST, among others, offering broad coverage of relevant cognitive dimensions.

### Methodological quality assessment

Methodological quality was evaluated across 11 dimensions: inclusion criteria (IC), random allocation (RA), allocation concealment (AH), baseline comparability, participant blinding, therapist blinding, outcome assessor blinding, dropout reporting (DR), intention-to-treat analysis, between-group comparisons (BT), and outcome measurement and variability. Each item was scored 0 or 1, with total scores categorized into four levels: excellent (≥8), good (6–7), fair (4–5), and poor (<4). Studies scoring below 4 were considered methodologically weak and excluded from this review. According to the scoring results (Supplemental Table S2, Supplemental Fig. S1), most studies demonstrated high methodological quality, with approximately one-third rated as "excellent" and over half achieving "good" or better. In summary, the included studies displayed generally robust methodological quality, though improvements remain necessary in bias prevention and the rigor of blinding protocols during intervention delivery.

### Quantitative findings

*Dose–response predictions of acute exercise on executive function in ADHD*. Before modeling the dose–response relationship, we conducted integrated evaluations of acute exercise effects on executive function in children and adolescents with ADHD using direct comparisons, indirect comparisons, and network meta-analysis. Results showed no significant differences across the three

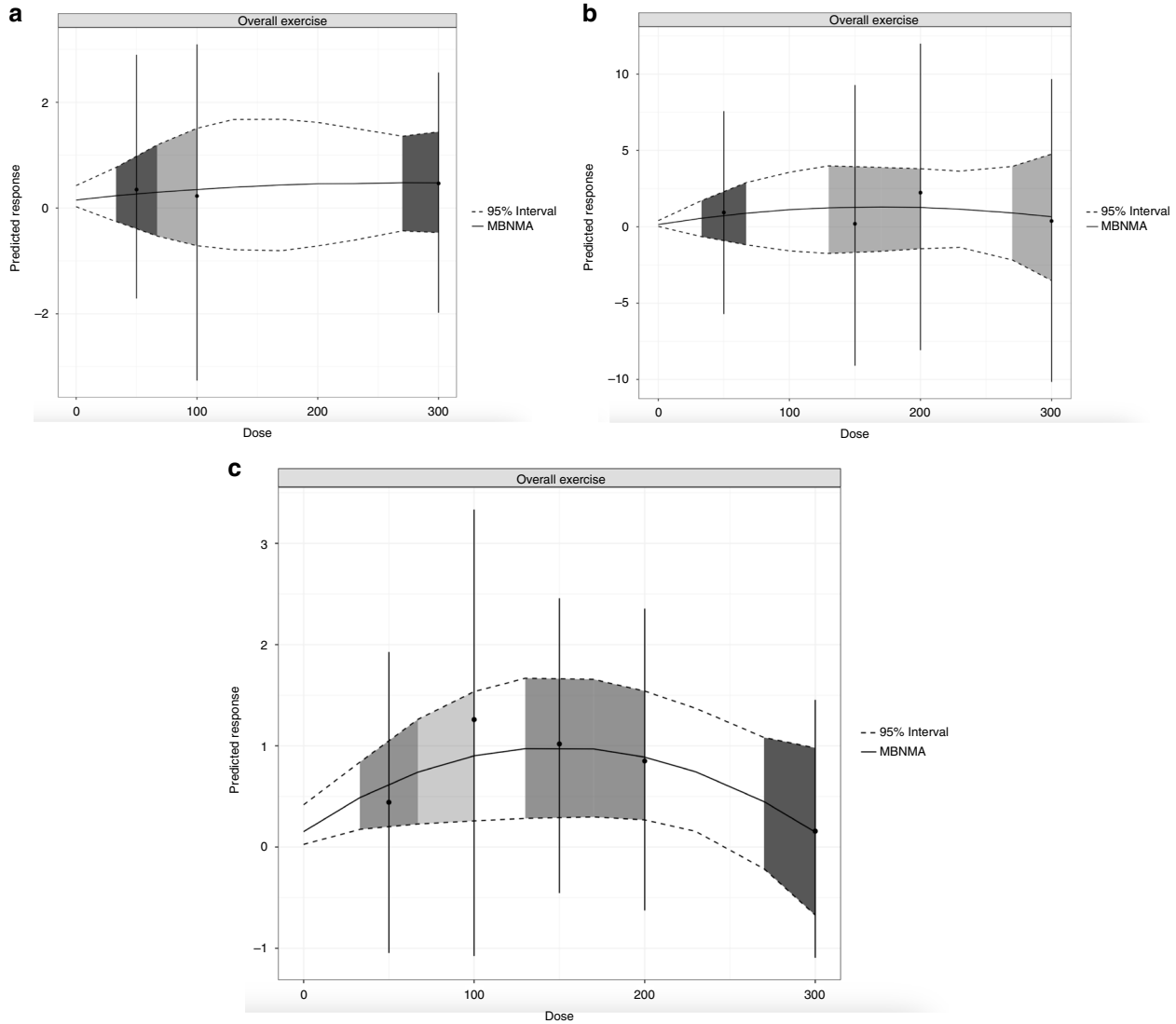



**Fig. 2 Acute Effects of Exercise on Executive Functions in ADHD.** Predicted dose–response relationships are shown for (**a**) cognitive flexibility, **b** working memory, and **c** inhibitory control, based on acute exercise interventions. Curves represent Bayesian model estimations with 95% credible intervals. The x-axis denotes standardized exercise dose (METs × min), and the y-axis represents standardized mean differences (SMD). Shaded areas indicate 95% credible intervals.

approaches, indicating high consistency and stability of findings across different evidence paths (Supplemental Table S5).

Subsequent dose–response analysis (Fig. 2) revealed nonlinear associations between acute exercise dose and improvements in each domain of executive function. The optimal predicted dose for cognitive flexibility was 270 METs, yielding an estimated SMD of 0.48 (95% CrI = −0.43, 1.36); for working memory, the optimal dose was 170 METs (SMD = 1.27, 95% CrI = −1.61, 3.89); and for inhibitory control, the peak benefit was observed at 130 METs (SMD = 1.27, 95% CrI = 0.28, 1.67). Among these, the CrIs for cognitive flexibility and working memory crossed zero, indicating that their effects were not statistically significant (Supplemental Table S6).

*Dose–response predictions of long-term exercise on executive function in ADHD.* Before modeling the dose–response curve for long-term interventions, we conducted pooled analyses using direct comparisons, indirect comparisons, and network meta-analysis. In the working memory domain, a significant difference was detected between intervention groups receiving 3000 METs and 1500 METs per week, suggesting a potential dose-dependent effect or nonlinearity that reduced statistical robustness. In contrast, no significant differences were observed across other dose combinations or executive function domains, indicating relatively consistent effects across conditions.

Further dose–response modeling (Fig. 3, Supplemental Figs. S2–5) indicated that cognitive flexibility reached its optimal predicted improvement at 1100 METs per week (SMD = 1.20, 95% CrI = 0.21, 2.09). Given the instability of working memory estimates at 3000 and 1500 METs, these doses were excluded from further interpretation to avoid bias. Instead, a reference dose of 1300 METs per week was selected, at which point the predicted improvement in working memory was most favorable (SMD = 0.82, 95% CrI = 0.08, 1.62). Inhibitory control showed an optimal predicted response at 2500 METs per week (SMD = 0.74, 95% CrI = 0.21, 1.29) (Supplemental Tables S7, 8).

Additionally, dose–response predictions by exercise type revealed distinct optimal doses for different domains. For cognitive flexibility, swimming at 1000 METs per week was most effective, followed by ball games at 1400 METs. For working memory, martial arts interventions at 500 METs per week yielded the highest effect, followed closely by rope skipping at 750 METs.

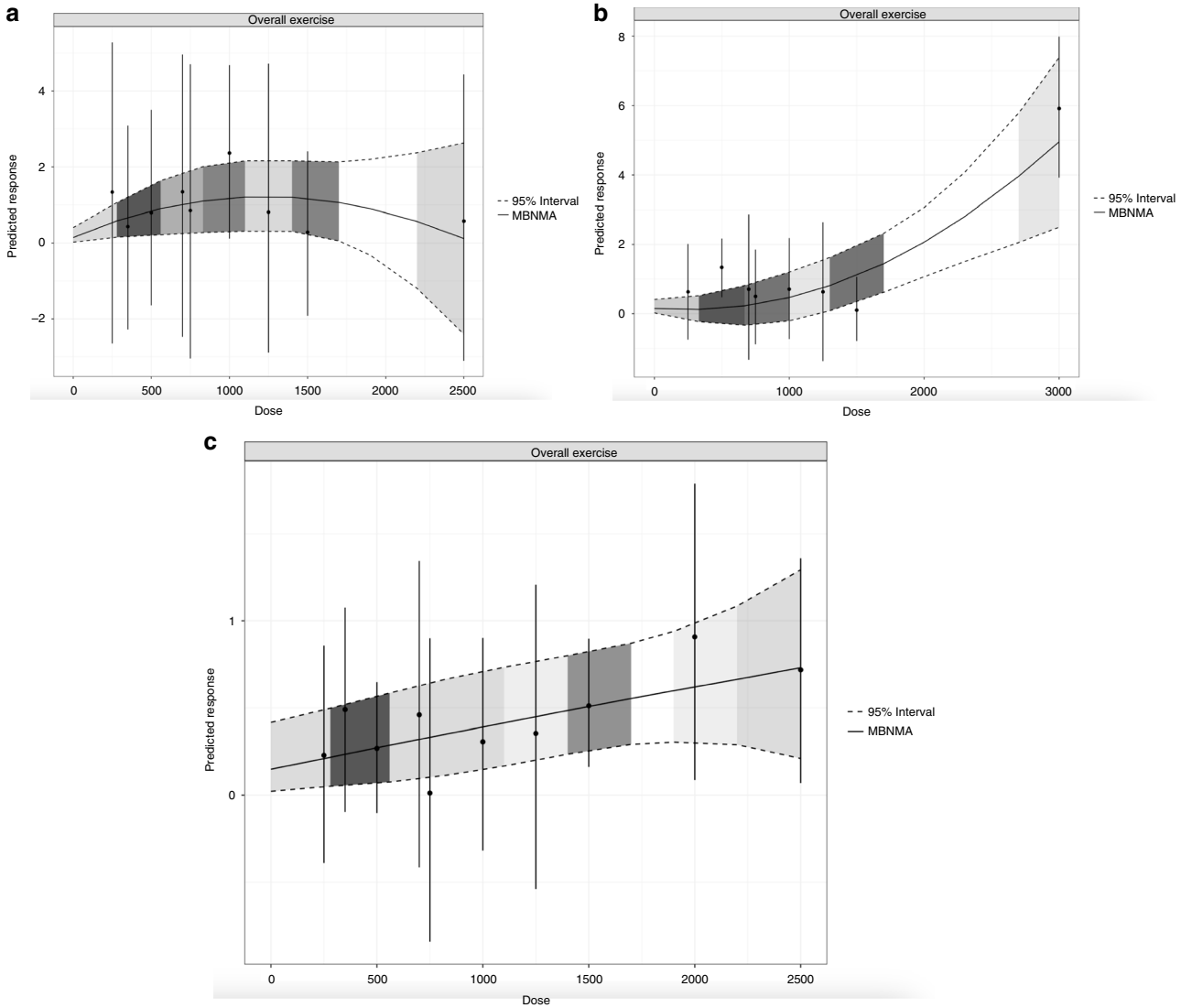



**Fig. 3 Long-Term Effects of Exercise on Executive Functions in ADHD.** Predicted dose–response relationships are shown for (**a**) cognitive flexibility, **b** working memory, and **c** inhibitory control, based on long-term exercise interventions. Bayesian model estimations are presented with 95% credible intervals. The x-axis shows standardized exercise dose, and the y-axis denotes SMD. Shaded areas indicate 95% credible intervals.

Inhibitory control showed the greatest improvement at 1000 METs per week of running, with a secondary peak at 830 METs (Supplemental Tables S9–12). These findings highlight the domain-specific nature of optimal exercise types and dosages, underscoring the importance of individualized intervention strategies. Detailed data for all dose–response predictions are provided in Supplemental Table S11.

*Quality of evidence assessment*. According to the GRADE evaluation, the 33 studies included in this review provide preliminary evidence supporting the positive effects of exercise interventions on executive function in children and adolescents with ADHD; however, the quality of evidence varies across executive function domains (Supplemental Tables S3, 4). For working memory, although the estimated effect size was relatively large, the limited number of studies and small sample sizes introduced considerable imprecision and risk of bias, resulting in a rating of low-quality evidence. For cognitive flexibility, the overall evidence was also rated as low quality, due to concerns about inconsistency and potential bias, despite acceptable precision and no clear additional benefit observed. In contrast, for inhibitory control, most studies were of high methodological quality, with consistent findings and significant effect sizes, leading to a rating of moderate-quality evidence.

Given that this review focuses on non-pharmacological interventions, which are generally safe and well-tolerated, and that the broader literature provides extensive support for the health benefits of exercise in youth populations, even domains with lower evidence ratings may still offer practical relevance. The presence of consistent positive trends and moderate-to-large effect sizes suggests that the main conclusions remain valid and that exercise interventions hold initial but meaningful promise for enhancing executive function in this population.

*Sensitivity analyses*. A series of sensitivity analyses were conducted to test the robustness of the dose–response model. First, various nonlinear functions were tested, and the quadratic model consistently demonstrated the best fit across all executive function domains, and was therefore retained for final modeling. Second, studies deemed at high risk of bias were excluded, and re-analysis using only high-quality studies yielded dose–response curves with similar effect patterns and optimal dose estimates,

suggesting that study quality did not significantly distort the model structure. Third, a leave-one-out analysis was performed to evaluate the influence of individual studies on the overall effect sizes and inflection points. No study was found to exert a dominant or unduly influential effect. Finally, studies involving extreme exercise doses (e.g., >2500 METs/week) were removed, and the model was re-run. The dose–response curve structure remained stable, with optimal dose estimates remaining within a reasonable range. Collectively, these results support the reliability and interpretability of the dose–response relationships modeled in this study.

*Bayesian meta-regression results*. Under chronic intervention conditions, age showed a statistically significant positive predictive effect on cognitive flexibility and inhibition, while a positive but non-significant trend was observed for working memory. Group sport demonstrated positive trends across all three domains, but none reached statistical significance. Under acute intervention conditions, both age and group sport exhibited positive but non-significant predictive effects in all three domains (Supplemental Table S13).

*Publication bias*. Assessment of publication bias in long-term exercise interventions revealed some asymmetry in the funnel plot for cognitive flexibility. Egger's test indicated potential bias ($p < 0.01$), suggesting that smaller studies may have been more likely to report inflated effect sizes (Supplemental Figs. S6, 7). However, sensitivity analyses showed that exclusion of potentially biased studies did not materially alter the main findings. Further correction using the trim-and-fill method confirmed that both the direction and magnitude of effects remained stable, indicating that publication bias had limited impact on the overall conclusions. For working memory and inhibitory control, funnel plots were more symmetrical and Egger's tests were non-significant ($p > 0.1$), indicating a low risk of publication bias in these domains. Overall, the findings of this review are statistically robust and further supported by theoretical frameworks and prior empirical literature, thereby enhancing confidence in the conclusions drawn.

## DISCUSSION

### First evidence of a dose–response relationship between exercise and executive function in ADHD

This study is distinct in that it is the first to systematically characterize the dose–response relationship between total exercise exposure and executive function in children and adolescents with ADHD, providing separate quantitative predictions for acute and long-term interventions across different executive subdomains. Unlike previous research that primarily treated exercise as a simple binary variable (intervention vs. control) or focused on comparing broad intervention types, our analysis integrates intensity, frequency, session duration, and overall program length into a standardized total dose framework. By doing so, this work fills a critical gap in the literature, offering essential dose–response evidence that has long been missing from this field. Our findings reveal a striking divergence between acute and long-term exercise interventions in improving executive function among children and adolescents with ADHD. For acute interventions, we observed distinctly nonlinear dose–response relationships: cognitive flexibility peaked at approximately 270 METs, while optimal effects on working memory and inhibitory control emerged at 170 METs and 130 METs, respectively. However, these acute effects warrant caution, as confidence intervals for cognitive flexibility and working memory were broad and crossed zero, indicating statistical uncertainty.[38]

In contrast, long-term exercise interventions produced more consistent and robust improvements. Optimal doses for cognitive flexibility, working memory, and inhibitory control increased markedly to around 1100 METs/week, 1300 METs/week, and 2500 METs/week, with narrower confidence intervals indicating reliable benefits. This pattern underscores the critical role of sustained, higher cumulative exercise exposure in achieving substantial executive function gains, particularly for inhibitory control, which appeared to demand the largest total dosage.[39] Beyond total dose, this study highlights the domain-specific advantages of particular exercise modalities. Swimming (1000 METs/week) and ball games (1400 METs/week) were most effective for enhancing cognitive flexibility, likely due to demands for spatial navigation, rapid task switching, and complex movement planning that engage frontoparietal networks under elevated metabolic loads. Martial arts (500 METs/week) and rope skipping (750 METs/week) produced the greatest improvements in working memory, perhaps by combining sequential action memory and dynamic updating at moderate intensities, optimizing dorsolateral prefrontal cortex activation without overtaxing executive resources. Meanwhile, running (830–1000 METs/week) yielded the strongest gains in inhibitory control, potentially reflecting enhanced functional coupling between the anterior cingulate cortex and ventral striatum through rhythmic motor activity that supports impulse control and error monitoring.[40]

It is noteworthy that in many of the trials included in this analysis, control groups exhibited relatively modest changes in executive function over the study periods. This may suggest that relying solely on natural development, everyday activities, or nonspecific exposure is often insufficient to produce substantial improvements. In contrast, structured exercise interventions generally demonstrated more pronounced benefits, indicating that well-designed exercise programs may offer a promising approach to supporting executive function in children and adolescents with ADHD.

These findings challenge the traditional notion of exercise as a uniformly generalized enhancer of executive function, instead demonstrating a dose-dependent, domain-specific, and modality-sensitive pattern of benefits. They suggest that future interventions should strategically align exercise type and dose with the unique cognitive load characteristics and neural susceptibilities of targeted executive deficits. Moreover, these results offer compelling hypotheses for future neuroimaging and multimodal physiological studies to investigate dose thresholds and saturation effects on brain network dynamics, paving the way toward a mechanistically informed, quantitatively optimized approach to exercise prescription in ADHD.

### Potential mechanisms by which exercise improves executive function in children and adolescents with ADHD

Exercise improves executive dysfunction in children and adolescents with ADHD through multilayered neural mechanisms, and this study is the first to reveal their dose dependence and domain specificity using dose–response modeling. First, acute intervention data showed that different executive subdomains reached peak effects at specific doses: approximately 270 METs for cognitive flexibility, 170 METs for working memory, and 130 METs for inhibitory control, all following a nonlinear pattern. These results suggest that improvements in executive function are not linearly proportional to exercise dose, but rather follow an "optimal dose window" in line with electrophysiological regulation principles. Previous research has shown that individuals with ADHD often exhibit elevated theta activity and reduced alpha rhythms, impairing attention maintenance and inhibitory control.[41–43] Short bouts of moderate-intensity exercise can enhance alpha activity and suppress theta rhythms within 30 min, enabling rapid modulation of executive performance.[44–46] Combined with the dose thresholds identified in this study, these findings suggest that low-to-moderate acute exercise may yield short-term cognitive benefits via immediate modulation of brain oscillations.

Second, dose–response analyses of chronic interventions revealed dose-linked features of neurobiological adaptation. Peak benefits were observed at 1100 METs/week for cognitive flexibility, 2500 METs/week for inhibitory control, and 1300 METs/week for working memory. These levels far exceed those required for short-term effects, suggesting that the mechanisms of improvement involve not only electrophysiological regulation but also long-term adaptation within neurotransmitter systems and brain structures. Regular physical activity has been shown to increase dopamine and norepinephrine release, enhancing attentional control and impulse regulation in individuals with ADHD.[47,48] In addition, high-dose exercise may activate neurotrophic factors such as BDNF, inducing neuroplastic remodeling of the prefrontal cortex, hippocampus, and basal ganglia, increasing cerebral blood flow, and enhancing connectivity within neural networks, ultimately improving cognitive function.[49–51] These long-term structural changes align with the cognitive benefits observed in the high-dose groups in this study, supporting the presence of a cumulative, dose-sensitive neuroregulatory mechanism.

Crucially, this study also revealed domain-specific matching between exercise types and executive functions: swimming (1000 METs/week) was most effective for enhancing cognitive flexibility, martial arts and rope skipping (500–750 METs/week) were optimal for improving working memory, while running (1000 METs/week) was most beneficial for inhibitory control. Though distinct in form, these activities share rhythmic features and cognitive demands, suggesting that the cognitive benefits of exercise are influenced not only by dose but also by brain activation patterns and cognitive loads inherent to the task.[52,53] The prefrontal cortex is more strongly activated in open-ended, dynamically adjusted exercise contexts, enhancing goal-directed behavioral control. Taken together, the results indicate that the beneficial effects of exercise on executive function in ADHD follow a distinct dose–domain–modality matching pattern. Acute cognitive benefits may arise from low-to-moderate doses through electrophysiological regulation, while medium-to-high chronic doses may drive neurochemical and structural adaptations. The dose–response model proposed in this study not only supports the existence of these neural mechanisms but also provides a precise, dose-informed basis for the clinical development of individualized exercise interventions.

### Comparison with and extension of previous research

The findings of this study are generally consistent with existing literature, while offering critical extensions in terms of dose–response relationships, exercise modality effects, and domain-specific cognitive outcomes. Most previous studies have focused on the average effects of exercise on executive function in individuals with ADHD, with limited efforts to systematically model and quantify dose–effect relationships.[13,52,54] Using network meta-analysis and nonlinear modeling, this study is the first to demonstrate that each executive function domain has its own optimal dose range, following a nonlinear pattern. This not only helps explain the inconsistencies observed in previous intervention studies but also highlights the central role of dose control in intervention design. Moreover, our results reveal domain-specific effects of different exercise types, supporting the concept of "task–cognition matching" as a basis for precision-based interventions.[53,55] In contrast to earlier studies that emphasized total intervention duration, we found that session length had limited influence on acute effects, suggesting that dosage parameters such as intensity and frequency may be more critical than time in determining intervention efficacy. Overall, by introducing "dose" into both intervention effect modeling and mechanistic explanation, this study advances the field toward a more precision-oriented approach to exercise-based interventions.

### Limitations and future directions

*Limitations in dose estimation*. Although this study is the first to construct a dose–response prediction model for executive function in children and adolescents with ADHD, most of the original studies did not report full METs data. As a result, exercise doses had to be estimated indirectly based on intervention intensity, frequency, and duration, which may have affected the accuracy of dose calculation. Future research should incorporate standardized reporting of exercise intensity and employ wearable devices for real-time monitoring of individual energy expenditure, thereby improving the precision and ecological validity of dose–effect modeling.

*Heterogeneity in interventions and participant characteristics*. Although the included studies involved children and adolescents with relatively consistent ADHD diagnoses (primarily based on DSM-IV criteria), the analyses lacked stratification by sex, ADHD subtypes, comorbid symptoms (e.g., anxiety, learning disorders), or socioeconomic background. This limits our ability to detect differential effects of exercise interventions across individual characteristics. Future studies should include more detailed subgroup analyses and covariate control to explore how individual differences influence responsiveness to exercise.

*Limited research on combined exercise–medication mechanisms and ecological translation*. Approximately one-third of the studies included in this review involved concurrent pharmacological treatment, but high-quality RCTs investigating the interactive mechanisms between exercise and medication are still lacking. In addition, most interventions were conducted in controlled laboratory environments, with limited assessment of ecological effectiveness in real-world settings such as school-based physical activity or home-based exercise. Future research should explore combined exercise–medication protocols using mechanistic designs and extend intervention studies to naturalistic settings to enhance the real-world applicability and sustainability of exercise strategies.

## CONCLUSION

Our findings clearly demonstrate that exercise interventions significantly improve executive function in children and adolescents with ADHD, with effects exhibiting a non-linear dependence on total dose and distinct optimal ranges and saturation points across cognitive subdomains. Acute and long-term interventions displayed markedly different dose–response profiles, underscoring the need to align dosing strategies with specific intervention durations and cognitive targets. Notably, our analysis revealed that swimming and ball games were most effective for enhancing cognitive flexibility at approximately 1000–1400 METs/week; martial arts and rope skipping yielded the greatest improvements in working memory at 500–750 METs/week; and running produced the strongest benefits for inhibitory control within the 830–1000 METs/week range. This domain-specific, modality-sensitive pattern of dose dependence highlights the necessity of moving beyond generalized recommendations toward individualized exercise prescriptions tailored to distinct executive deficits. These insights not only provide a quantitative basis for elucidating the neurocognitive mechanisms by which exercise ameliorates executive dysfunction in ADHD, but also establish critical foundations for developing evidence-based, dose-controlled intervention protocols.

## DATA AVAILABILITY

This study is a systematic review and network meta-analysis and does not involve the generation of new data.

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## AUTHOR CONTRIBUTIONS

Q. Pan drafted the manuscript and performed the primary data analysis. Pengwei He conceived and supervised the study and critically revised the manuscript. S. Zheng contributed to data collection and interpretation.

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## COMPETING INTERESTS

The authors declare no competing interests.

## ETHICAL APPROVAL AND INFORMED CONSENT

All authors confirm that they are aware of and comply with ethical publication practices and consent to the submission of this manuscript.

## ADDITIONAL INFORMATION

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